WORKED SOLUTION

STEP III 1989, Question 10

Pure • Riemann Sums

STEP III 1989, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q10

This question has a full guided treatment in *Beyond STEP*, Section 4.1, *What Is an Integral?*: https://beyondstep.keithmcnulty.org/ch04/riemann.html

The key idea

A product of consecutive integers is the "falling power" version of r^5, and it sums by telescoping, exactly as r^5 integrates: the difference of two consecutive six-term products is six times a five-term product. Parts (i) and (ii) give a product sum just above and just below $\sum r^5$. Part (iii) then squeezes the two Riemann sums of x^5 between these bounds, and both bounds tend to the same limit.

Solution

Part (i)

Write $P(r) = r(r+1)(r+2)(r+3)(r+4)(r+5)$. Then $P(r)$ and $P(r-1) = (r-1)r(r+1)(r+2)(r+3)(r+4)$ share the five factors $r(r+1)\cdots(r+4)$, so

$$\begin{aligned} P(r) - P(r-1) &= r(r+1)(r+2)(r+3)(r+4)\big[(r+5) - (r-1)\big] \\ &= 6\,r(r+1)(r+2)(r+3)(r+4). \end{aligned}$$

Summing from $r = 1$ to n, the right-hand side telescopes:

$$\begin{aligned} \sum_{r=1}^{n} r(r+1)(r+2)(r+3)(r+4) &= \tfrac{1}{6}\sum_{r=1}^{n}\big[P(r) - P(r-1)\big] = \tfrac{1}{6}\big[P(n) - P(0)\big] \\ &= \tfrac{1}{6}\,n(n+1)(n+2)(n+3)(n+4)(n+5), \end{aligned}$$

since $P(0) = 0$. (Induction on n works equally well; the inductive step is the same factorisation.)

For the deduction: if $r \geq 1$ then each factor satisfies $r + k > r > 0$ for $k = 1, 2, 3, 4$, so

$$r^5 < r(r+1)(r+2)(r+3)(r+4) \qquad (r \geq 1).$$

Adding these strict inequalities for $r = 1, \ldots, n$ gives $\sum_{r=1}^{n} r^5 < \frac{1}{6}n(n+1)(n+2)(n+3)(n+4)(n+5)$, as required.

Part (ii)

Now we want a product that sits *below* r^5, so we shift the factors down: use $(r-4)(r-3)(r-2)(r-1)r$. With $Q(r) = (r-4)(r-3)(r-2)(r-1)r(r+1)$, the same factorisation gives

$$\begin{aligned} Q(r) - Q(r-1) &= (r-4)(r-3)(r-2)(r-1)r\big[(r+1) - (r-5)\big] \\ &= 6\,(r-4)(r-3)(r-2)(r-1)r. \end{aligned}$$

Summing from $r = 0$ to $n-1$:

$$\begin{aligned} \sum_{r=0}^{n-1} (r-4)(r-3)(r-2)(r-1)r &= \tfrac{1}{6}\big[Q(n-1) - Q(-1)\big] \\ &= \tfrac{1}{6}\,(n-5)(n-4)(n-3)(n-2)(n-1)n, \end{aligned}$$

because $Q(-1) = (-5)(-4)(-3)(-2)(-1)\cdot 0 = 0$.

Ask yourself Is $(r-4)(r-3)(r-2)(r-1)r < r^5$ true for every r in the sum?

Not quite, and this is where $n > 1$ comes in. We compare term by term:

- $r = 0$: both sides are 0, so we only have equality.
- $r = 1, 2, 3, 4$: the product contains a zero factor, so it is $0 < r^5$.
- $r \geq 5$: every factor satisfies $0 < r - k < r$ for $k = 1, 2, 3, 4$, so the product is $< r^5$.

So $(r-4)(r-3)(r-2)(r-1)r \leq r^5$ for all $r \geq 0$, with strict inequality for every $r \geq 1$. If $n > 1$, the sum from $r = 0$ to $n-1$ includes the term $r = 1$, so at least one inequality is strict, and

$$\sum_{r=0}^{n-1} r^5 > \tfrac{1}{6}(n-5)(n-4)(n-3)(n-2)(n-1)n.$$

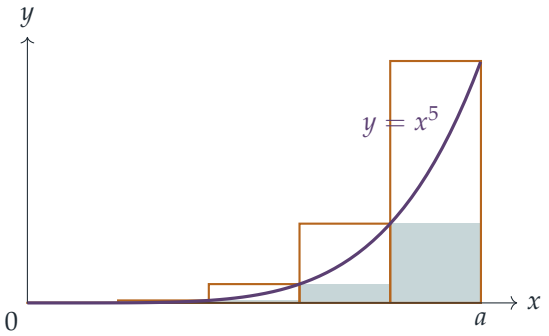
(For $n = 1$ both sides are 0, which is why the condition is needed.)

Part (iii)

Take $f(x) = x^5$, which is increasing. The two sums in the definition are

$$L_n = \sum_{r=0}^{n-1} \frac{a}{n}\Big(\frac{ra}{n}\Big)^5 = \frac{a^6}{n^6}\sum_{r=0}^{n-1} r^5, \qquad U_n = \sum_{r=1}^{n} \frac{a}{n}\Big(\frac{ra}{n}\Big)^5 = \frac{a^6}{n^6}\sum_{r=1}^{n} r^5.$$

For $a > 0$ these are the areas of the rectangles below and above the curve:



(Shaded: L_n; outlined: U_n; here $n = 5$.) The algebra below does not need $a > 0$, though: $a^6 > 0$ for any $a \neq 0$, and $a = 0$ is trivial since every sum is 0.

Using part (ii) for L_n (with $n > 1$) and part (i) for U_n,

$$L_n > \frac{a^6}{6n^6}(n-5)(n-4)(n-3)(n-2)(n-1)n = \frac{a^6}{6}\prod_{k=1}^{5}\left(1-\frac{k}{n}\right),$$

$$U_n < \frac{a^6}{6n^6}n(n+1)(n+2)(n+3)(n+4)(n+5) = \frac{a^6}{6}\prod_{k=1}^{5}\left(1+\frac{k}{n}\right).$$

Also $U_n - L_n = \dfrac{a^6}{n^6}\left(n^5 - 0^5\right) = \dfrac{a^6}{n} > 0$, so altogether

$$\frac{a^6}{6}\prod_{k=1}^{5}\left(1-\frac{k}{n}\right) < L_n < U_n < \frac{a^6}{6}\prod_{k=1}^{5}\left(1+\frac{k}{n}\right).$$

As $n \to \infty$, each bracket tends to 1, so both outer bounds tend to $\frac{1}{6}a^6$. By the squeeze (sandwich) theorem, L_n and U_n both converge, and to the same limit $\frac{1}{6}a^6$. So both limits in the definition exist and are equal, and therefore

$$\int_0^a x^5 \,\mathrm{d}x = \tfrac{1}{6}a^6.$$

Checks

- Small cases of (i): for $n = 1$, $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120 = \frac{1}{6} \cdot 720$; for $n = 2$, $120 + 720 = 840 = \frac{1}{6} \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7$.
- The identities in (i) and (ii), and both inequalities, hold exactly for every n up to 200 (checked by computer).
- With $a = 1$ and $n = 1000$: $L_n \approx 0.166167$ and $U_n \approx 0.167167$, either side of $\frac{1}{6} \approx 0.166667$, and $U_n - L_n = 0.001 = a^6/n$.

Common slips

- Claiming $(r-4)\cdots r < r^5$ for all $r \geq 0$: at $r = 0$ it is an equality, so the strict inequality needs the $r = 1$ term, that is, $n > 1$.
- Showing only that L_n and U_n are bounded, or only that $U_n - L_n \to 0$. The definition needs both limits to *exist*; the squeeze gives existence and the value together.

Beyond STEP

Part (iii) is a hands-on version of the Riemann (or Darboux) integral: a function is integrable when its lower and upper sums can be squeezed together, and for a monotonic function the gap is always $(b-a)\,|\mathrm{f}(b) - \mathrm{f}(a)|/n$, so every monotonic function on a closed interval is integrable. The products in (i) and (ii) are "rising" and "falling factorial" powers, which make the calculus of finite differences look just like ordinary calculus. Abbott's *Understanding Analysis* develops the integral this way.

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